

- 7 Many devices are designed to create a spray of tiny droplets. The effectiveness of these devices usually depends on droplet size. One example is an agriculture pesticide spray in which a few large droplets do not coat the leaves of plants as well as many small droplets. Another example is a fuel injection system for an engine.

Measuring the size of droplets present in a spray is difficult to do by direct means but instruments called droplet sizers can be purchased which make droplet sizing a fast routine operation.

The principle of operation of one such sizer is shown in Fig. 7.1, in which light from a helium/neon laser is passed through a spray of droplets of uniform diameter and forms a circular diffraction ring of radius x . The diameter d of the droplets is related to x by the equation

$$d = k \frac{\lambda}{x}$$

In this equation λ , the wavelength of the light, is 6.33×10^{-7} m. k is a constant equal to 0.474 m, and d and x are both in metres.

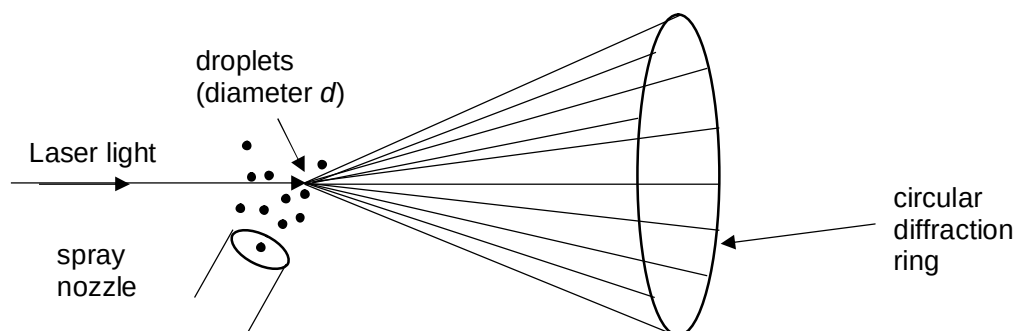


Fig. 7.1

In practice, a spray will consist of droplets of different sizes, so many rings of diffracted light will be caused. The diffraction pattern in Fig. 7.2a is projected on a flat surface containing many light sensitive detectors. The output from the detectors can be analysed by a computer and be shown in the form of a graph in Fig. 7.2b.

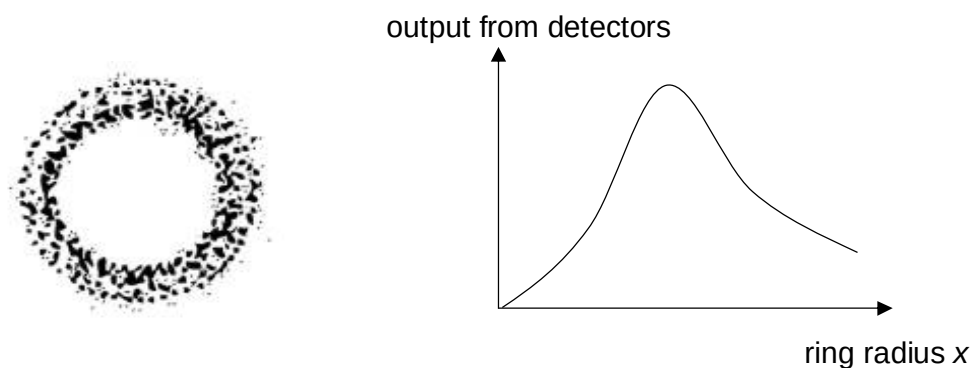


Fig. 7.2a

Fig. 7.2b

- (a) Suggest two devices, other than those mentioned in the first paragraph of the passage, where droplet size is important.

Spray paint [B1], Hair spray [B1]

[2]

- (b) Outline a direct method for measuring droplet diameter.
Apparatus available includes: high speed camera, stroboscope, stopwatch, rulers.

Use high speed camera to capture still image of droplets. [1]
Have a backdrop of grid squares in the background to estimate size of each droplet. [1]
Use an object of known diameter at same distance from background to compare with size of droplet and thus use scaling to determine droplet diameter. [1]

[3]

- (c) Give two reasons why direct methods are likely to be difficult for measuring droplets of small diameter.

1. Measurement devices may lack sufficient precision for measurement. [B1]
2. The droplets may be too packed to give a distinct image [B1]

[2]

- (d) Calculate the value of x for a droplet of diameter

- (i) $10\ \mu\text{m}$

$$d = k \frac{\lambda}{x}$$

$$x = k \frac{\lambda}{d} = 0.474 \left(\frac{6.33 \times 10^{-7}}{10 \times 10^{-6}} \right) = 0.0300\ \text{m} \quad [\text{A1}]$$

$x = \dots\dots\dots\ \text{m} \quad [1]$

- (ii) $200\ \mu\text{m}$

$$d = k \frac{\lambda}{x}$$

$$x = k \frac{\lambda}{d} = 0.474 \left(\frac{6.33 \times 10^{-7}}{200 \times 10^{-6}} \right) = 0.00150\ \text{m} \quad [\text{A1}]$$

$x = \dots\dots\dots\ \text{m} \quad [1]$

- (e) State whether a small value of x corresponds to large or to small droplets.

Small x corresponds with large droplets [1]

[1]

- (f) Suggest, with reference to the equation given, how the radius of the circular diffraction ring can be increased for the same diameter of water droplets.

Increase wavelength of laser light. [1]

[1]

- (g) Sketch on the axes in Fig. 7.3 curves to show the general shape of graphs that would be obtained if

- (i) droplets with a wide range of diameter were used. Label as (i). [1]

- (ii) very small droplets with a narrow range of diameters were used. Label as (ii). [1]

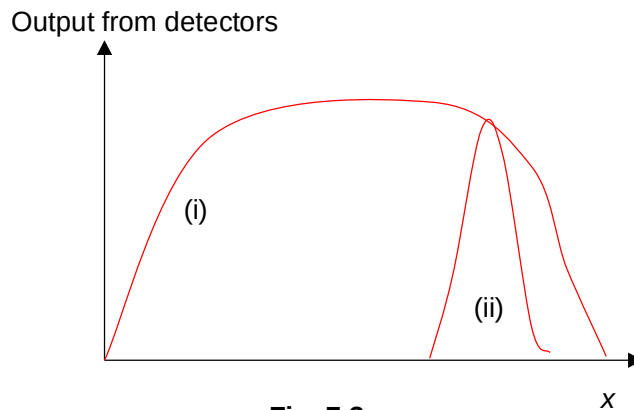


Fig. 7.3

- (h) In practice, a cloud of spray droplets moves through the laser beam as shown at intervals in Fig. 7.4. The output from the detectors varies with time in the way shown in Fig. 7.5.

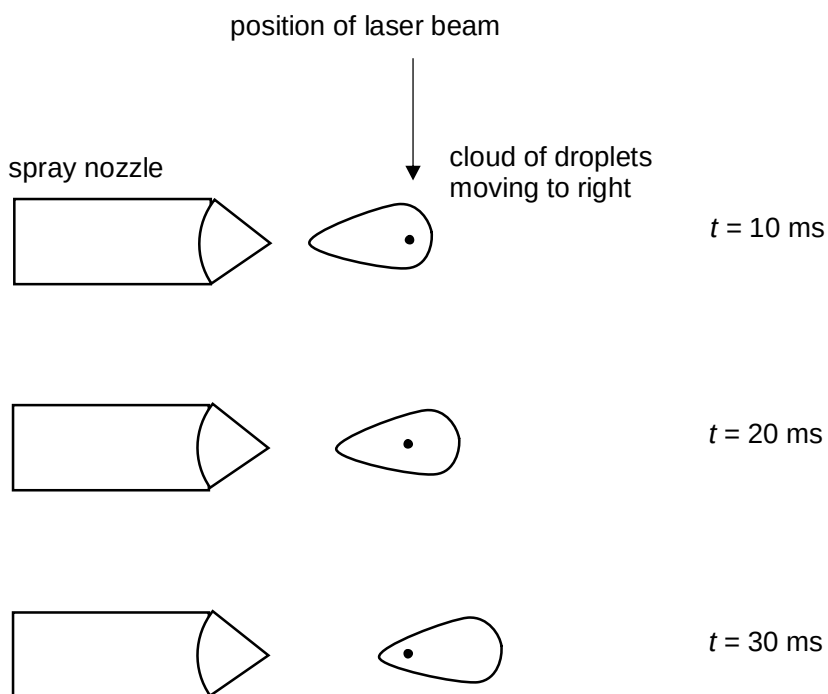


Fig. 7.4

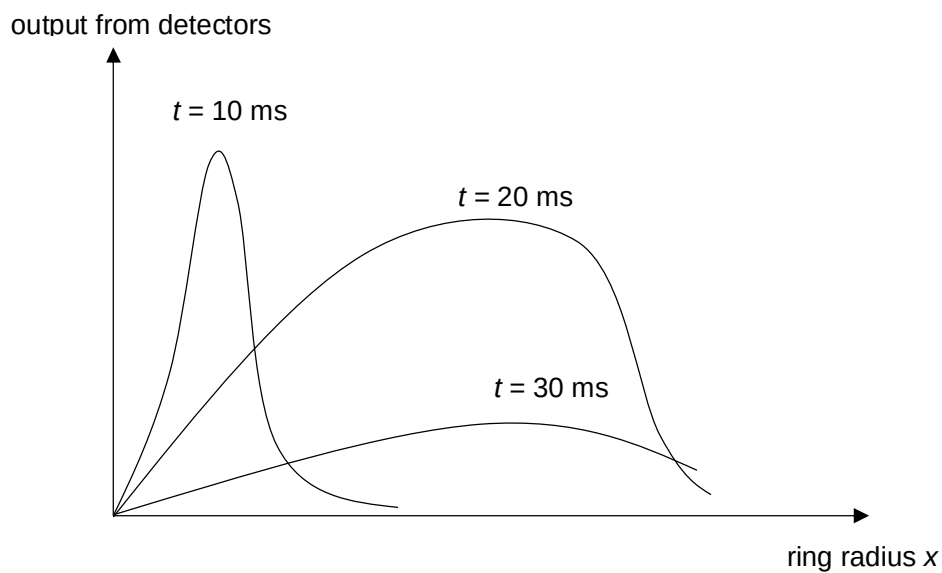


Fig. 7.5

With reference to Fig. 7.6, describe the distribution of droplets in the cloud according to their size and concentration (i.e. amount of droplets)..

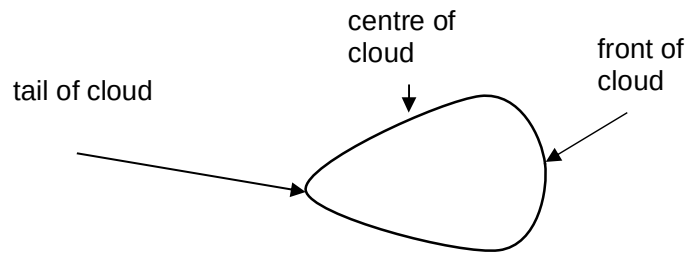


Fig. 7.6

At the front tip of the cloud,
Droplets are of uniform sizes and with a large diameter. [1] fewer droplets as compared to centre [1]

At the centre of cloud,
Droplets are of a range of sizes [1] and are large number of droplets as compared to front and tail of cloud. [1]

At tail of cloud, detected with a range of different diameters. [1] fewer droplets as compared to centre [1]

Concentration – amount of water droplets
Size – diameter + range of diameters

[6]

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